

CAFÉ LANDING PAGE

A Mini Project Report

submitted by

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to the APJ Abdul Kalam Technological University
in partial fulfilment of the requirements for the award of the Degree
of
Master of Computer Applications



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Declaration

I undersigned hereby declare that the project report “CAFÉ LANDING PAGE” submitted for partial fulfilment of the requirements for the award of degree of Master of Computer Applications of the APJ Abdul Kalam Technological University, Kerala, is a bonafide work done by me under supervision of Mr. Nowshad CV, Assistant Professor, Department of Computer Applications. This submission represents my ideas in my own words and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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07/10/2026

DEPARTMENT OF COMPUTER APPLICATIONS
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CERTIFICATE

This is to certify that the report entitled **CAFÉ LANDING PAGE** is a bonafide record of the Mini Project work during the year 2026-27 carried out by **KADHEEJA NAISHA (MES25MCA-2029)** submitted to the APJ Abdul Kalam Technological University, in partial fulfilment of the requirements for the award of the Master of Computer Applications, under my guidance and supervision. This report in any form has not been submitted to any other University or Institution for any purpose.

Internal Supervisor

Head of The Department

Acknowledgment

I express my sincere gratitude to our principal Dr. Rahumathunisa I for permitting and providing all facilities to carry out this project. I would like to express my sincere gratitude to my internal supervisor and product owner, Mr. Nowshad CV, Assistant Professor, for the invaluable guidance, support, and encouragement throughout the development of this mini project. I also extend my thanks to the Head of the Department Prof. Hyderali K and all faculty members of the Department of Computer Applications, MES College of Engineering, for providing the necessary resources and a conducive environment for learning and innovation.

I am grateful to the café owners and friends who shared how they present their businesses online. Their experiences shaped the sections and features of this landing page.

Finally, I thank my family and friends for their unwavering support and motivation. Above all, I extend my heartfelt gratitude to the Almighty for giving me the strength and wisdom to complete this project.

Kadheeja Naisha
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Abstract

Café Landing Page is a single-page website that gives a local coffee shop, *Kettle & Crumb*, a warm and inviting presence on the web. Many independent cafés depend only on word of mouth and social media, where the menu, opening hours and address are scattered, outdated or missing. This project brings everything a visitor needs onto one scrollable page: a full-width hero with the café's name and tagline, a menu of signature drinks, food and daily specials with prices, the weekly opening hours, an embedded Google Map with the address, and an Instagram-style photo gallery.

The site is built with HTML5, CSS3 and vanilla JavaScript. Menu items are stored as JSON data and rendered dynamically, and a small password-protected admin panel lets the owner add products, change prices, mark daily specials and remove items without touching the code. Changes are persisted in the browser's LocalStorage and the site is deployed on GitHub Pages. The project followed the Agile Scrum methodology over six sprints spanning 41 days. This report outlines the motivation, objectives, system design, implementation and results of the project, culminating in a lightweight, responsive and easily maintainable web presence for a small business.

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Chapter 1. Introduction

A café is more than a place to buy coffee—it is a space people choose for its atmosphere, its food and the people behind the counter. Before visiting, most customers now search online to see what a café serves, when it is open and how to get there. Yet many independent cafés still have no website of their own and rely entirely on word of mouth and social media profiles, where essential information is scattered, incomplete or out of date.

Café Landing Page addresses this gap by giving a local café, *Kettle & Crumb* in Fort Kochi, a single, attractive online home. Everything a visitor needs is presented on one scrollable page, organised into clear sections: hero, menu, opening hours, location map and gallery. A small admin panel allows the owner to add new products and update prices without editing any code.

The project falls under the domain of front-end web development, with a focus on visual design, typography, responsive layout and usability. The page is built with HTML5, CSS3 and a small amount of vanilla JavaScript, which keeps it lightweight, quick to load, cheap to host and easy to maintain.

The problem addressed is the lack of a dedicated, owner-controlled web presence for small cafés. Existing options are either generic social profiles that give little control over branding and layout, or full website builders that are costly and complex for a small business. A focused landing page offers an affordable middle path.

1.1 Motivation

Customers expect to find a café's menu, hours and location online instantly, and increasingly they do so on a mobile phone. A strong first impression on a small screen can decide whether someone walks in. Many independent cafés, however, have no website, and the little information available about them is spread across different platforms.

A single landing page is an affordable marketing asset that a small business can own and control. The problem is concrete, visual and solvable with front-end skills, which made it an ideal mini project: it demonstrates an eye for design alongside coding ability, and the result is something a real café could use from day one.

1.2 Objectives

The primary objectives of the *Café Landing Page* project are:

- To give a local café a single, attractive online home for its essential information.
- To present the menu, opening hours and location clearly on one scrollable page.
- To create a strong first impression through warm design, typography and imagery.
- To deliver a fully responsive layout that works on desktop, tablet and mobile.
- To let the owner add new products and update prices through a simple admin panel.
- To keep the site lightweight so it is quick to build, cheap to host and easy to maintain.

1.3 Contributions

Key contributions of the *Café Landing Page* include:

- A responsive single-page website with hero, menu, opening hours, location and gallery sections.
- A menu that is stored as JSON data and rendered dynamically with JavaScript, grouped by category.
- A live “Open now” status calculated from the weekly opening hours.
- An embedded Google Map with address, contact details and a directions link.
- A password-protected admin panel to add, edit, delete and mark items as daily specials, with changes saved to browser LocalStorage.

These contributions address the limitations of existing approaches and give a small café a professional, easily maintained web presence.

1.4 Report Organization

The report is organized as follows. Chapter 2 gives an overview of the existing system and the proposed solution along with its functionalities. Chapter 3 describes the development methodology, the tools used, the modules designed, the Scrum artefacts and the data design. Chapter 4 presents the results with screenshots of the implemented user interface. Finally, Chapter 5 summarises the project's achievements and the future scope. The appendices contain the data flow diagrams, the ER diagram, selected source code and additional screenshots.

Chapter 2. System Study

The *Café Landing Page* is designed to give a small café a clear, owner-controlled presence on the web. This chapter studies how cafés currently share their information and describes the proposed system.

2.1 Existing System

Presently, most independent cafés rely only on word of mouth and social media. Their menu, opening hours and address are scattered across different platforms or are missing altogether. Social media profiles give little control over branding and layout, and information posted there is often outdated and hard to locate on a map. There is no dedicated space to convey the atmosphere of the café, and where a basic website does exist, prices and new items can only be updated by editing the code. The main drawbacks are:

- Menu, hours and address are scattered or missing.
- Little control over branding, layout and presentation.
- Information is often outdated and hard to find on a map.
- No dedicated space to show the café's atmosphere.
- Updating prices or adding items requires editing code.

Improving upon these limitations, the proposed system offers a single, well-structured page that the owner can keep up to date on their own.

2.2 Proposed System

The proposed system is a single, thoughtfully structured landing page built with HTML5, CSS3 and JavaScript. It offers:

- A full-width hero with the café's name, tagline, feature photograph and call-to-action buttons.
- A clear menu of signature drinks and daily specials with descriptions and prices.
- Opening hours and an embedded Google Maps location in one place.
- An Instagram-style photo gallery that conveys the atmosphere of the space.
- A small admin panel to add products and update prices.

Technologically, the menu is stored as JSON data and rendered by JavaScript; admin changes are saved in browser LocalStorage, the map is embedded with the Google Maps

Embed API, and the site is hosted on GitHub Pages. The design emphasises a warm aesthetic, good typography and generous imagery, with a responsive layout that adapts cleanly from desktop to mobile and smooth scrolling between sections.

Target users are café visitors—both new and regular customers looking for the menu, hours and location—and the café owner, who maintains the menu through the admin panel.

2.3 Functionalities of Proposed System

- **Hero:** full-width banner with the café name, tagline, photo and call-to-action.
- **Menu / Specials:** signature drinks and daily specials with descriptions and prices.
- **Opening Hours:** the week's schedule laid out clearly at a glance.
- **Location:** embedded Google Maps iframe together with the café's address.
- **Photo Gallery:** Instagram-style strip of images showing the atmosphere.
- **Responsive Design:** layout adapts cleanly from desktop to mobile.
- **Interactivity:** smooth scrolling, mobile navigation toggle and gallery behaviour.
- **Admin Panel:** password-protected page to add new products and edit prices.

Chapter 3. Methodology

3.1 Introduction

The project was developed using the Agile Scrum methodology. Requirements were captured as user stories in a product backlog, and the work was divided into short sprints, each with a clear sprint goal. One section of the page was designed, built and tested per sprint. Progress was reviewed with the product owner, Mr. Nowshad CV, after every sprint, and feedback was folded into the next sprint until the page was complete. Six sprints were carried out between 06/08/2026 and 15/09/2026.

3.2 Software Tools

Category	Tool / Technology
Operating System	Windows
Front End	HTML5, CSS3, JavaScript (Vanilla)
Menu Data	JSON file rendered dynamically with JavaScript
Data Persistence	Browser LocalStorage (admin changes)
Maps Integration	Google Maps Embed API
IDE / Code Editor	Visual Studio Code
Version Control	Git & GitHub
Deployment	GitHub Pages

Table 3.1: Software tools and languages used for project development

3.3 Module Description

The system is divided into five modules, each built in its own sprint, to keep it maintainable and easy to test.

3.3.1 Hero Section Module

This module creates the first impression of the café. It contains a full-width banner image, the café name and logo, a short tagline and a feature photograph. A primary call-to-action button (“See the menu”) and a secondary action (“Find us”) lead visitors further down the page. A sticky navigation bar stays at the top while scrolling, and links scroll smoothly to each section.

3.3.2 Menu Section Module

This module lists signature drinks, daily specials, and food and pastry items, each with a short description and price. Items are grouped by category (Coffee, Tea, Pastries and Food) and displayed in a responsive card grid. The menu is rendered from JSON data by JavaScript, so it always reflects the latest changes made in the admin panel.

3.3.3 Opening Hours & Location Module

This module shows the weekly opening hours, including weekend timings, with the current day highlighted. It displays the café address, an embedded Google Maps iframe in a responsive container, a “Get directions” link, the contact number and email, and social media links.

3.3.4 Gallery & Footer Module

This module presents an Instagram-style photo strip in a responsive image grid. Images are lazy-loaded and carry alt text for accessibility, with hover effects for interactivity. The footer provides quick links to every section, the café's address and contact, copyright and credits, and a back-to-top button.

3.3.5 Admin Panel Module

This module lets the café owner manage the menu. It provides a password-protected login, a form to add a new product, and options to edit the name, description and price, delete or hide an item, and mark an item as a daily special. The menu table gives a live preview, changes are saved to browser storage, and the menu data can be exported and imported as JSON.

3.4 User Story

ID	As a type of User	I want to	So that I can
1	Visitor	view the hero section	get a quick first impression of the café
2	Visitor	browse the menu	see drinks, specials and prices before visiting
3	Visitor	check the opening hours	know when the café is open
4	Visitor	view the location map	find directions to the café easily
5	Visitor	view the photo gallery	get a feel for the café's atmosphere
6	Admin	log into the admin panel	securely manage the menu
7	Admin	add a new product	keep the menu up to date

ID	As a type of User	I want to	So that I can
8	Admin	edit a product's price or details	reflect current offerings accurately
9	Admin	delete or hide an item	remove discontinued products

Table 3.2: User Stories

3.5 Product Backlog

ID	Name	Priority	Estimate (hrs)	Status
1	Design hero banner layout	High	3	Completed
2	Add café name, logo & tagline	High	2	Completed
3	Implement CTA buttons & sticky nav	Medium	3	Completed
4	Create menu JSON data structure	High	3	Completed
5	Render menu cards dynamically	High	4	Completed
6	Category grouping & responsive grid	Medium	3	Completed
7	Build opening hours table	Medium	2	Completed
8	Embed Google Maps location	High	3	Completed
9	Add contact & social links	Low	2	Completed
10	Build photo gallery grid	Medium	3	Completed
11	Add lazy loading & hover effects	Low	3	Completed
12	Design footer with quick links	Low	2	Completed
13	Password-protected admin login	High	3	Completed
14	Add / Edit / Delete product form	High	5	Completed
15	Save & sync changes to LocalStorage	High	4	Completed

Table 3.3: Product Backlog

3.6 Sprint Backlog

Backlog Item	Completion Date	Est. Hrs	D 1	D 2	D 3	D 4	D 5	Status
SPRINT 1 – HERO SECTION (06/08/2026 – 12/08/2026)								
Layout & wireframe planning	07/08/2026	4	2	2	0	0	0	Completed
Hero HTML/CSS + navigation bar	09/08/2026	6	0	2	2	2	0	Completed

Backlog Item	Completion Date	Est. Hrs	D 1	D 2	D 3	D 4	D 5	Status
Smooth scroll JS + responsive fixes	12/08/2026	5	0	0	1	2	2	Completed
SPRINT 2 – MENU SECTION (13/08/2026 – 19/08/2026)								
Menu JSON data structure	14/08/2026	3	2	1	0	0	0	Completed
Render menu cards from JSON	17/08/2026	5	0	1	2	2	0	Completed
Category grouping & responsive grid	19/08/2026	4	0	0	1	2	1	Completed
SPRINT 3 – OPENING HOURS & LOCATION (20/08/2026 – 26/08/2026)								
Opening hours table design	21/08/2026	3	2	1	0	0	0	Completed
Embed Google Maps + address	24/08/2026	4	0	1	2	1	0	Completed
Contact & social links section	26/08/2026	2	0	0	0	1	1	Completed
SPRINT 4 – GALLERY & FOOTER (27/08/2026 – 02/09/2026)								
Photo gallery grid layout	28/08/2026	4	2	2	0	0	0	Completed
Lazy loading & hover effects	31/08/2026	3	0	1	2	0	0	Completed
Footer, quick links & back-to-top	02/09/2026	3	0	0	1	1	1	Completed
SPRINT 5 – ADMIN PANEL (03/09/2026 – 09/09/2026)								
Password-protected admin login	04/09/2026	4	2	2	0	0	0	Completed
Add / Edit / Delete product form	07/09/2026	6	0	2	2	2	0	Completed
Save to LocalStorage + live preview	09/09/2026	4	0	0	1	2	1	Completed
SPRINT 6 – TESTING & DEPLOYMENT (10/09/2026 – 15/09/2026)								
Cross-browser & responsive testing	12/09/2026	4	2	2	0	0	0	Completed
Bug fixes & performance polish	14/09/2026	3	0	1	2	0	0	Completed
Final review & GitHub Pages deploy	15/09/2026	2	0	0	0	1	1	Completed

Table 3.4: Sprint Backlog

3.7 Project Plan

Sprint	Focus Area	Start Date	End Date	Days	Status
Sprint 1	Hero Section & Navigation	06/08/2026	12/08/2026	7	Completed
Sprint 2	Menu / Specials Section	13/08/2026	19/08/2026	7	Completed
Sprint 3	Opening Hours & Location	20/08/2026	26/08/2026	7	Completed
Sprint 4	Gallery & Footer	27/08/2026	02/09/2026	7	Completed
Sprint 5	Admin Panel	03/09/2026	09/09/2026	7	Completed
Sprint 6	Testing & Deployment	10/09/2026	15/09/2026	6	Completed
Total Project Duration				41	

Table 3.5: Project Plan

3.8 Database Design

The project does not use a relational database. The data model is implemented as a JSON schema: the default menu ships as a JSON file, and any changes made by the owner in the admin panel are persisted to the browser's LocalStorage under a single key. The schema has three entities—Category, Product and Admin—shown in the ER diagram in Appendix B. Each product belongs to one category, and the admin manages all products.

Field Name	Data Type	Description
category_id	String	Unique identifier of the category
name	String	Category name (Coffee, Tea, Pastries, Food)

Table 3.6: Category Schema

Field Name	Data Type	Description
product_id	String	Unique identifier of the product
name	String	Name of the menu item
description	String	Short one-line description
price	Number	Price in Indian Rupees (stored as a number, shown as ₹120)
category	String	Category the item belongs to (refers to Category)
image	String (URL)	Optional image URL for the item
isSpecial	Boolean	True if the item is a daily special

Table 3.7: Product Schema

Field Name	Data Type	Description
admin_id	String	Identifier of the café owner account

Field Name	Data Type	Description
username	String	Admin username
password	String	Admin password used to unlock the admin panel

Table 3.8: Admin Schema

The menu is seeded with twelve items across four categories, as listed below.

No.	Product Name	Category	Price	Daily Special
1	Cold brew	Coffee	₹160	No
2	Cortado	Coffee	₹140	Yes
3	Filter of the day	Coffee	₹120	No
4	Flat white	Coffee	₹150	No
5	Masala chai	Tea	₹90	No
6	First flush Darjeeling	Tea	₹130	Yes
7	Kashmiri kahwa	Tea	₹150	No
8	Sourdough cardamom bun	Pastries	₹110	Yes
9	Butter croissant	Pastries	₹130	No
10	Almond frangipane tart	Pastries	₹160	No
11	Whipped ricotta & fig toast	Food	₹220	Yes
12	Mushroom melt	Food	₹240	No

Table 3.9: Sample Menu Data

Chapter 4. Results and Discussions

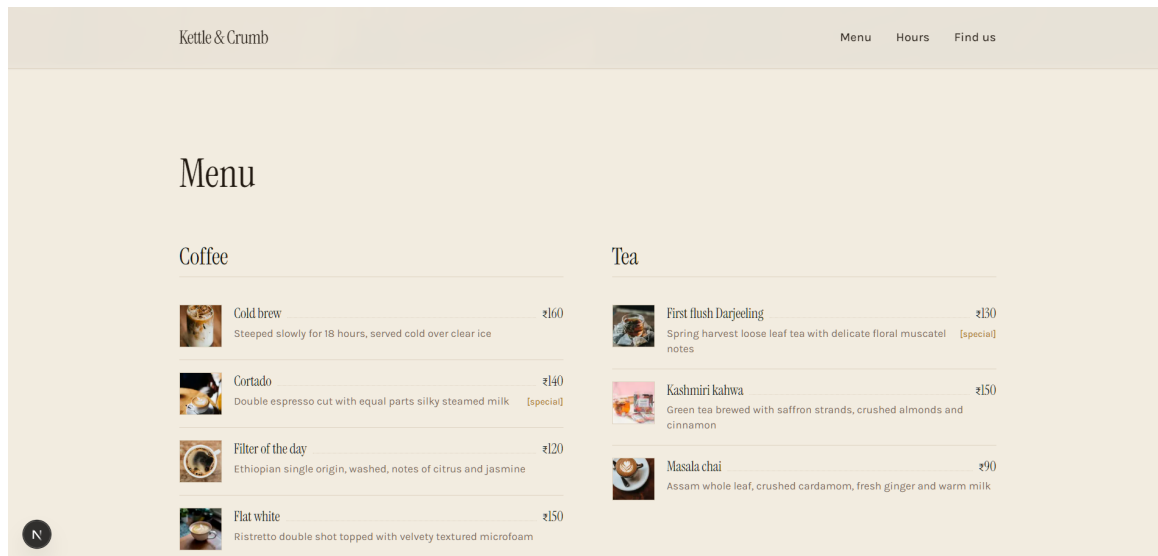
4.1 Results

The *Café Landing Page* was successfully developed and tested across all its modules on desktop, tablet and mobile screen sizes. Below are key screenshots and descriptions of the implemented features.

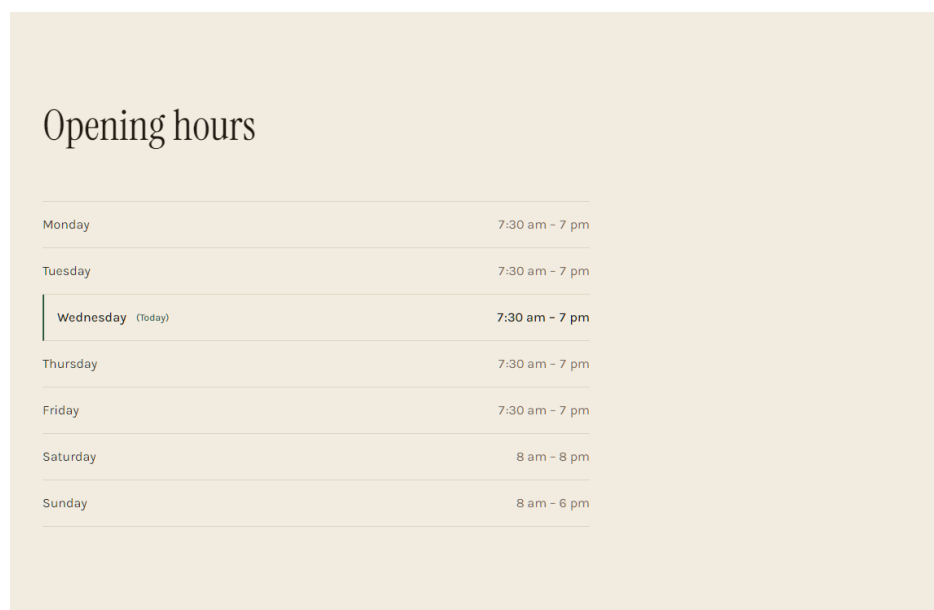


Figure 4.1: Hero Section

The hero section is the first thing a visitor sees. It shows the café name *Kettle & Crumb* over a warm full-width photograph, the tagline “Slow coffee and warm bread, since 2019.”, a live “Open now” indicator calculated from the opening hours, and two calls to action: “See the menu” and “Find us”. The sticky navigation bar links to the Menu, Hours and Find us sections.

**Figure 4.2: Menu Section**

The menu is rendered from JSON data and grouped by category. Each item shows a thumbnail, name, short description and price in rupees (for example, Cold brew at ₹160). Items marked by the owner as daily specials carry a *[special]* tag.

**Figure 4.3: Opening Hours Section**

The opening hours section lists the full week at a glance, with weekday hours of 7:30 am – 7 pm and separate weekend timings. The current day is highlighted so visitors can see today's hours instantly.

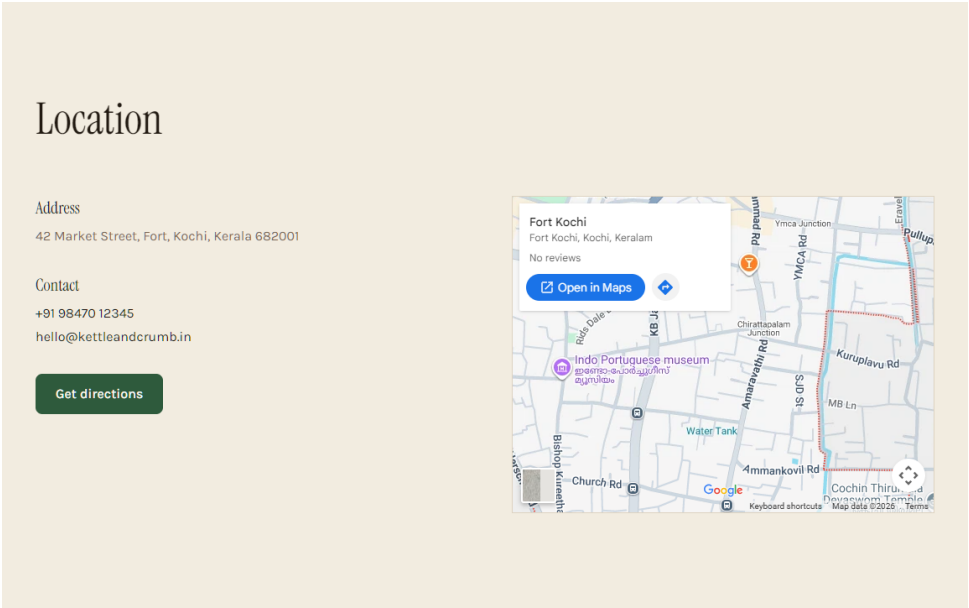


Figure 4.4: Location Section

The location section shows the café address, phone number and email next to an embedded Google Map of Fort Kochi. The “Get directions” button opens the route in Google Maps.



Figure 4.5: Photo Gallery Section

The gallery presents an Instagram-style grid of lazy-loaded photographs that convey the café's atmosphere. Every image carries descriptive alt text for accessibility.

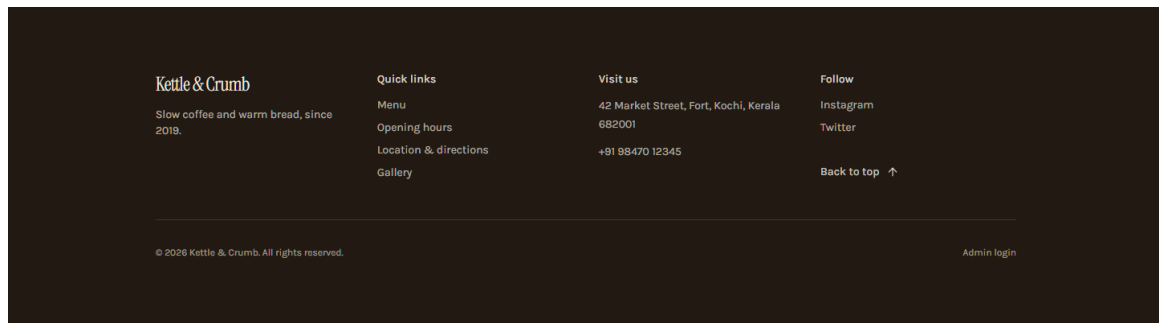


Figure 4.6: Footer

The footer provides quick links to every section, the café's address and phone number, social links, a back-to-top button and a discreet link to the admin login.

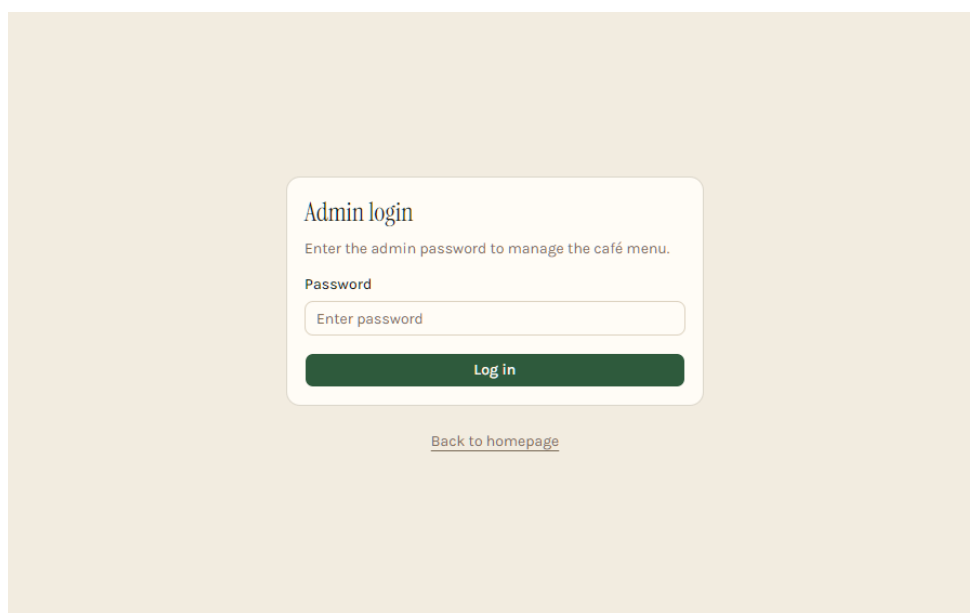
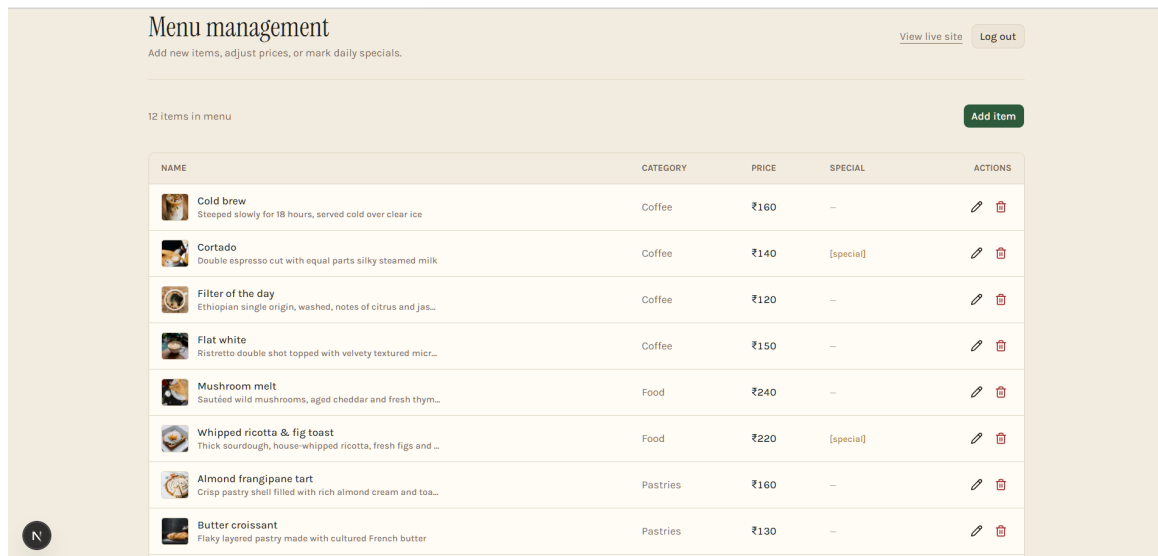


Figure 4.7: Admin Login Page

The admin panel is protected by a password. Only the café owner, who knows the password, can open the menu management page.



Menu management

Add new items, adjust prices, or mark daily specials.

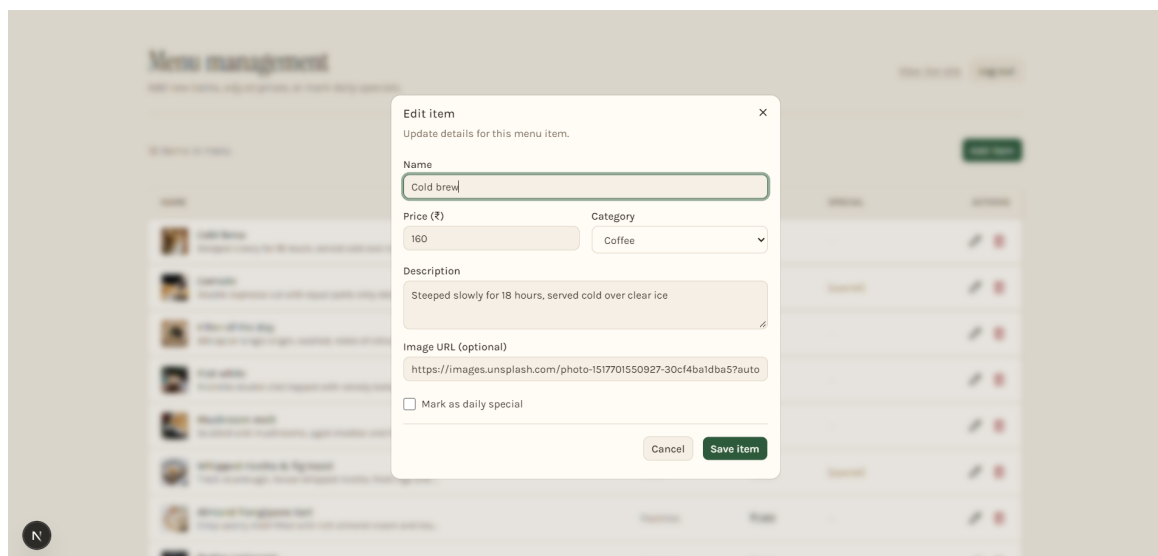
View live site Log out

12 items in menu Add item

NAME	CATEGORY	PRICE	SPECIAL	ACTIONS
 Cold brew Steeped slowly for 18 hours, served cold over clear ice	Coffee	₹160	—	 
 Cortado Double espresso cut with equal parts silky steamed milk	Coffee	₹140	[special]	 
 Filter of the day Ethiopian single origin, washed, notes of citrus and jas...	Coffee	₹120	—	 
 Flat white Ristretto double shot topped with velvety textured micr...	Coffee	₹150	—	 
 Mushroom melt Sautéed wild mushrooms, aged cheddar and fresh thym...	Food	₹240	—	 
 Whipped ricotta & fig toast Thick sourdough, house-whipped ricotta, fresh figs and ...	Food	₹220	[special]	 
 Almond frangipane tart Crisp pastry shell filled with rich almond cream and toa...	Pastries	₹160	—	 
 Butter croissant Flaky layered pastry made with cultured French butter	Pastries	₹130	—	 

Figure 4.8: Menu Management Dashboard

After logging in, the owner sees all menu items in a table with name, category, price and special status. The “Add item” button opens a form for a new product, and each row has edit and delete actions.



Menu management

Add new items, adjust prices, or mark daily specials.

View live site Log out

12 items in menu Add item

NAME

CATEGORY

PRICE

SPECIAL

ACTIONS

 **Cold brew**
Steeped slowly for 18 hours, served cold over clear ice

Coffee

₹160

—

 **Cortado**
Double espresso cut with equal parts silky steamed milk

Coffee

₹140

[special]

 **Filter of the day**
Ethiopian single origin, washed, notes of citrus and jas...

Coffee

₹120

—

 **Flat white**
Ristretto double shot topped with velvety textured micr...

Coffee

₹150

—

 **Mushroom melt**
Sautéed wild mushrooms, aged cheddar and fresh thym...

Food

₹240

—

 **Whipped ricotta & fig toast**
Thick sourdough, house-whipped ricotta, fresh figs and ...

Food

₹220

[special]

 **Almond frangipane tart**
Crisp pastry shell filled with rich almond cream and toa...

Pastries

₹160

—

 **Butter croissant**
Flaky layered pastry made with cultured French butter

Pastries

₹130

—

Edit item

Update details for this menu item.

Name

Cold brew

Price (₹)

160

Category

Coffee

Description

Steeped slowly for 18 hours, served cold over clear ice

Image URL (optional)

https://images.unsplash.com/photo-1517701550927-30c4ba1dba57?auto

☐ Mark as daily special

Cancel Save item

Figure 4.9: Edit Item Dialog

The edit dialog lets the owner update the name, price, category, description and image URL of an item, and mark it as a daily special. Saving the item updates the stored menu and the public page re-renders with the new data.

Chapter 5. Conclusion and Future Scope

The *Café Landing Page* successfully gives a local café a single, attractive and owner-controlled home on the web. It brings the menu, opening hours, location and atmosphere of the café together on one responsive page, and lets the owner keep the menu up to date without touching any code.

Key achievements include:

- A warm, responsive single-page design that works on desktop, tablet and mobile.
- A dynamic menu rendered from JSON data, grouped by category and highlighting daily specials.
- Opening hours with a live “Open now” status and an embedded Google Maps location.
- A password-protected admin panel to add, edit and delete products and update prices.
- Delivery within six Agile Scrum sprints over 41 days, with every backlog item completed.

The project demonstrates how a small set of front-end technologies can solve a real, everyday problem for a small business in a lightweight and affordable way.

Future enhancements could include online ordering and table reservations, a customer reviews section, a shared online database so that menu changes made on one device appear on all devices, multiple admin accounts with stronger authentication, image upload instead of image URLs, and support for regional languages such as Malayalam.

The main limitation of the current version is that admin changes are stored in the browser's LocalStorage, so they live only on the device where they were made, and the single shared password offers basic rather than strong security. These can be addressed in future iterations with a backend service and proper user accounts.

References

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Appendix

Appendix A Data Flow Diagram

LEVEL 0 – CONTEXT DIAGRAM

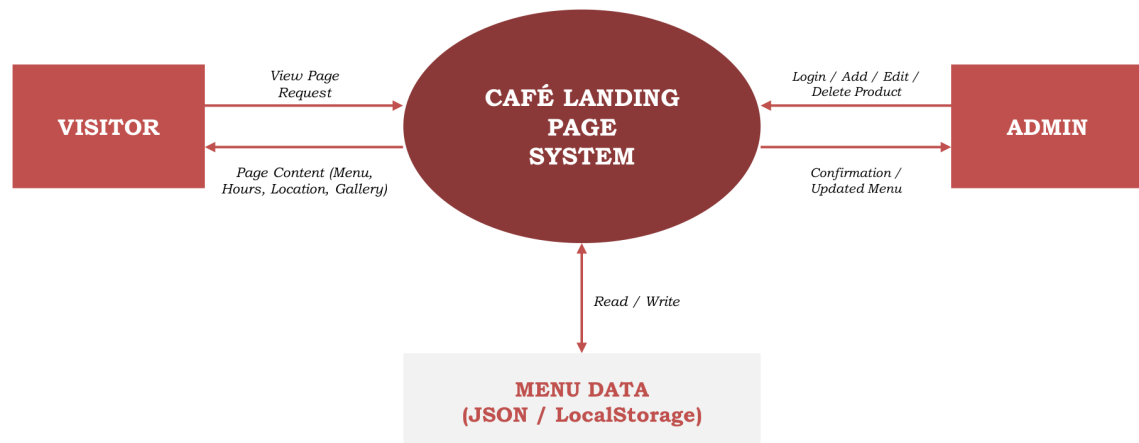


Figure A.1: Level 0 – Context Diagram

LEVEL 1 – PROCESS BREAKDOWN

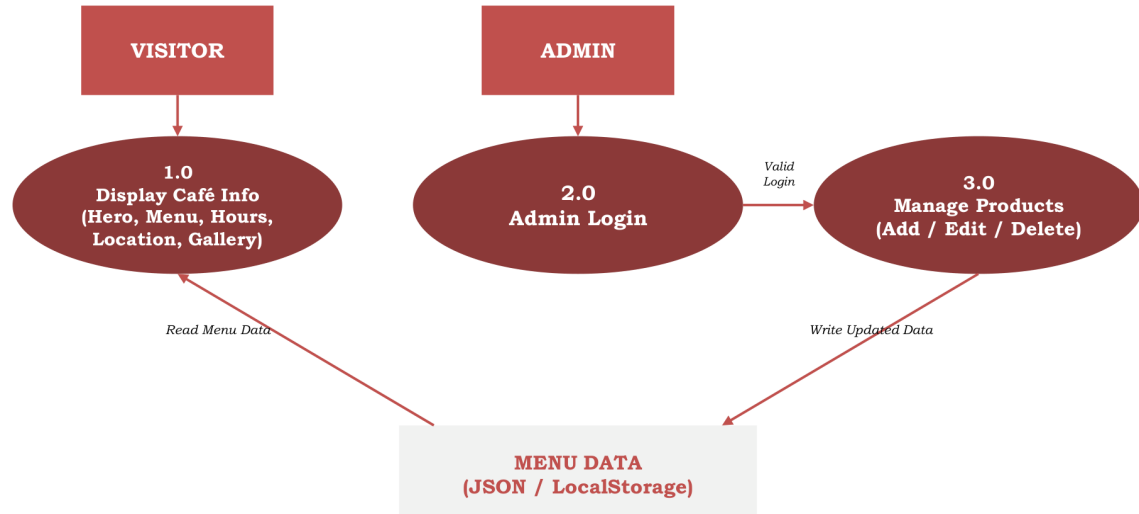


Figure A.2: Level 1 – Process Breakdown

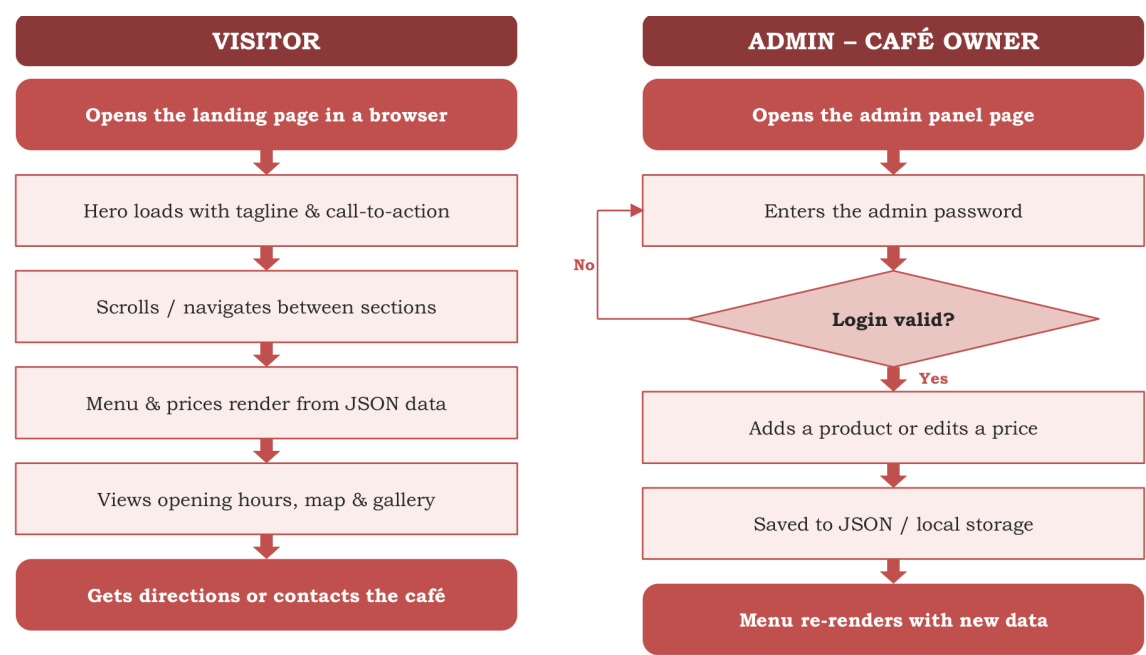
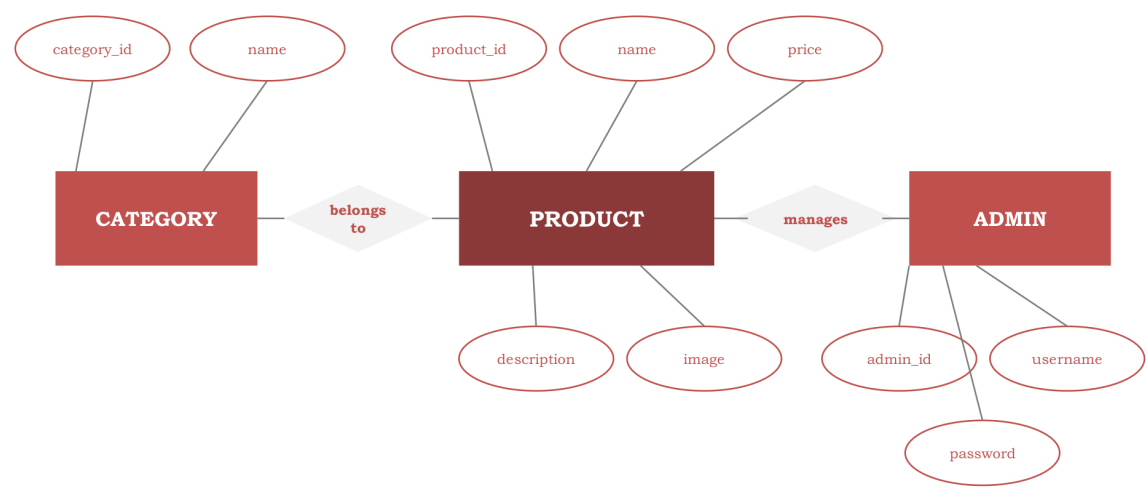


Figure A.3: Workflow of Visitor and Admin

Appendix B ER Diagram



Note: implemented as a JSON schema, persisted to browser LocalStorage (no relational database).

Figure B.1: ER Diagram

Appendix C Source Code

1. Hero Section (index.html)

```
<header class="navbar" id="navbar">
  <a href="#top" class="logo">Kettle & Crumb</a>
  <button class="nav-toggle" aria-label="Open menu" aria-expanded="false">&#9776;</button>
  <nav class="nav-links">
    <a href="#menu">Menu</a>
    <a href="#hours">Hours</a>
    <a href="#location">Find us</a>
  </nav>
</header>

<section class="hero" id="top">
  <div class="hero-content">
    <h1>Kettle & Crumb</h1>
    <p class="tagline">Slow coffee and warm bread, since 2019.</p>
    <p class="open-status"><span class="dot"></span><span id="openStatus"></span></p>
    <div class="hero-actions">
      <a href="#menu" class="btn btn-primary">See the menu</a>
      <a href="#location" class="btn-link">Find us</a>
    </div>
  </div>
</section>
```

2. Menu Data (menu.json)

```
[
  { "id": "p1", "name": "Cold brew", "category": "Coffee", "price": 160,
    "description": "Steeped slowly for 18 hours, served cold over clear ice",
    "image": "images/cold-brew.jpg", "isSpecial": false },
  { "id": "p2", "name": "Cortado", "category": "Coffee", "price": 140,
    "description": "Double espresso cut with equal parts silky steamed milk",
    "image": "images/cortado.jpg", "isSpecial": true },
  { "id": "p5", "name": "Masala chai", "category": "Tea", "price": 90,
    "description": "Assam whole leaf, crushed cardamom, fresh ginger and warm milk",
    "image": "images/masala-chai.jpg", "isSpecial": false }
]
```

3. Rendering the Menu from JSON (script.js)

```
const STORAGE_KEY = "cafeMenu";
const CATEGORY_ORDER = ["Coffee", "Tea", "Pastries", "Food"];

async function loadMenu() {
  const saved = localStorage.getItem(STORAGE_KEY);
  if (saved) return JSON.parse(saved);
  try {
    const response = await fetch("data/menu.json");
    return await response.json();
  } catch (error) {
    document.getElementById("menuGrid").textContent =
      "Sorry, the menu could not be loaded right now.";
    return [];
  }
}

function formatPrice(price) {
  return "\u20B9" + price;
}

function renderMenu(products) {
  const grid = document.getElementById("menuGrid");
  grid.innerHTML = "";
  CATEGORY_ORDER.forEach((category) => {
    const items = products.filter((p) => p.category === category);
    if (items.length === 0) return;
    const group = document.createElement("div");
    group.className = "menu-group";
    group.innerHTML = `<h3>${category}</h3>` + items.map((item) => `
      <article class="menu-item">
        
        <div class="menu-text">
          <div class="menu-row">
            <h4>${item.name}</h4>
            <span class="price">${formatPrice(item.price)}</span>
          </div>
          <p>${item.description}
            ${item.isSpecial ? '<span class="special">[special]</span>' : ''}</p>
        </div>
      </article>`).join("");
    grid.appendChild(group);
  });
}

loadMenu().then(renderMenu);
```

4. Opening Hours and Open-Now Status

```
const HOURS = [
  { day: "Sunday", open: "08:00", close: "18:00" },
  { day: "Monday", open: "07:30", close: "19:00" },
  { day: "Tuesday", open: "07:30", close: "19:00" },
  { day: "Wednesday", open: "07:30", close: "19:00" },
  { day: "Thursday", open: "07:30", close: "19:00" },
  { day: "Friday", open: "07:30", close: "19:00" },
  { day: "Saturday", open: "08:00", close: "20:00" }
];

function toMinutes(time) {
  const [hours, minutes] = time.split(":").map(Number);
  return hours * 60 + minutes;
}

function updateOpenStatus() {
  const now = new Date();
  const today = HOURS[now.getDay()];
  const current = now.getHours() * 60 + now.getMinutes();
  const isOpen = current >= toMinutes(today.open) && current < toMinutes(today.close);
  const closeHour = Number(today.close.split(":")[0]) % 12 || 12;
  document.getElementById("openStatus").textContent = isOpen
    ? `Open now \u2014 until ${closeHour} pm`
    : "Closed now";
  document.querySelector(`[data-day="${today.day}"]`).classList.add("today");
}

updateOpenStatus();
```

5. Location Section with Google Maps

```
<section class="location" id="location">
  <h2>Location</h2>
  <div class="location-grid">
    <div class="location-info">
      <h4>Address</h4>
      <p>42 Market Street, Fort, Kochi, Kerala 682001</p>
      <h4>Contact</h4>
      <p>+91 98470 12345<br>hello@kettleandcrumb.in</p>
      <a class="btn btn-primary" target="_blank" rel="noopener"
        href="https://maps.google.com/?q=Fort+Kochi+Kerala">Get directions</a>
    </div>
    <div class="map-container">
      <iframe title="Map showing Kettle & Crumb in Fort Kochi" loading="lazy"
        src="https://www.google.com/maps/embed?pb=..."
        referrerpolicy="no-referrer-when-downgrade"></iframe>
    </div>
  </div>
</section>
```

6. Smooth Scroll, Mobile Navigation and Back-to-Top

```
const toggle = document.querySelector(".nav-toggle");
const navLinks = document.querySelector(".nav-links");

toggle.addEventListener("click", () => {
  const isOpen = navLinks.classList.toggle("open");
  toggle.setAttribute("aria-expanded", isOpen);
});

document.querySelectorAll('a[href^="#"]').forEach((link) => {
  link.addEventListener("click", (event) => {
    const target = document.querySelector(link.getAttribute("href"));
    if (!target) return;
    event.preventDefault();
    target.scrollIntoView({ behavior: "smooth" });
    navLinks.classList.remove("open");
  });
});

const backToTop = document.getElementById("backToTop");
window.addEventListener("scroll", () => {
  backToTop.classList.toggle("visible", window.scrollY > 600);
});
backToTop.addEventListener("click", () => window.scrollTo({ top: 0, behavior: "smooth" }));
```

7. Admin Login (admin.js)

```
const ADMIN_PASSWORD = "*****"; // set by the café owner before deployment

document.getElementById("loginForm").addEventListener("submit", (event) => {
  event.preventDefault();
  const password = document.getElementById("password").value;
  const error = document.getElementById("loginError");
  if (password === ADMIN_PASSWORD) {
    sessionStorage.setItem("isAdmin", "true");
    showDashboard();
  } else {
    error.textContent = "That password is not correct.";
  }
});

function showDashboard() {
  document.getElementById("loginView").hidden = true;
  document.getElementById("dashboard").hidden = false;
  renderTable();
}

if (sessionStorage.getItem("isAdmin") === "true") showDashboard();
```

8. Add, Edit and Delete Products with LocalStorage

```
let products = JSON.parse(localStorage.getItem("cafeMenu")) || [];

function saveProducts() {
  localStorage.setItem("cafeMenu", JSON.stringify(products));
  renderTable();
}

function handleSave(event) {
  event.preventDefault();
  const form = event.target;
  const product = {
    id: form.dataset.editId || "p" + Date.now(),
    name: form.name.value.trim(),
    price: Number(form.price.value),
    category: form.category.value,
    description: form.description.value.trim(),
    image: form.image.value.trim(),
    isSpecial: form.isSpecial.checked
  };
  if (!product.name || product.price <= 0) {
    alert("Please enter a name and a valid price.");
    return;
  }
  const index = products.findIndex((p) => p.id === product.id);
  if (index >= 0) products[index] = product;
  else products.push(product);
  saveProducts();
  closeDialog();
}

function deleteProduct(id) {
  products = products.filter((p) => p.id !== id);
  saveProducts();
}

function renderTable() {
  document.getElementById("itemCount").textContent = `${products.length} items in menu`;
  document.getElementById("productRows").innerHTML = products.map((p) => `
    <tr>
      <td>${p.name}<small>${p.description}</small></td>
      <td>${p.category}</td>
      <td>\u20B9${p.price}</td>
      <td>${p.isSpecial ? "[special]" : "\u2013"}</td>
      <td><button onclick="openDialog('${p.id}')" aria-label="Edit">Edit</button>
        <button onclick="deleteProduct('${p.id}')" aria-label="Delete">Delete</button></td>
    </tr>`
  ).join("");
}
```

Appendix D Screenshots

1. Full Menu (all categories)

Menu	
Coffee	
	Cold brew Steeped slowly for 18 hours, served cold over clear ice ₹160
	Cortado Double espresso cut with equal parts silky steamed milk [special] ₹140
	Filter of the day Ethiopian single origin, washed, notes of citrus and jasmine ₹120
	Flat white Ristretto double shot topped with velvety textured microfoam ₹150
Tea	
	First flush Darjeeling Spring harvest loose leaf tea with delicate floral muscatel notes [special] ₹130
	Kashmiri kahwa Green tea brewed with saffron strands, crushed almonds and cinnamon ₹150
	Masala chai Assam whole leaf, crushed cardamom, fresh ginger and warm milk ₹90
Pastries	
	Almond frangipane tart Crisp pastry shell filled with rich almond cream and toasted flakes ₹160
	Butter croissant Flaky layered pastry made with cultured French butter ₹130
	Sourdough cardamom bun Twisted Swedish-style bun with freshly ground green cardamom [special] ₹110
Food	
	Mushroom melt Sautéed wild mushrooms, aged cheddar and fresh thyme on sourdough ₹240
	Whipped ricotta & fig toast Thick sourdough, house-whipped ricotta, fresh figs and wild honey [special] ₹220

Figure D.1: Full Menu with All Categories

2. Mobile View

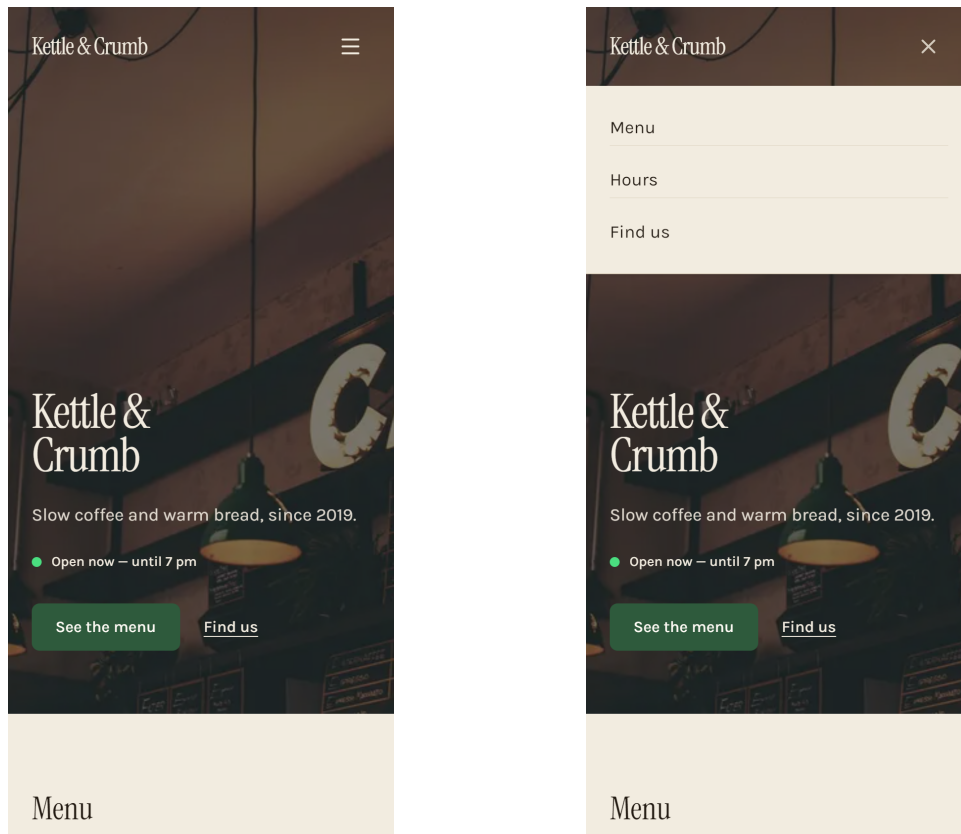


Figure D.2: Mobile View – Hero and Navigation Menu

3. Mobile Menu

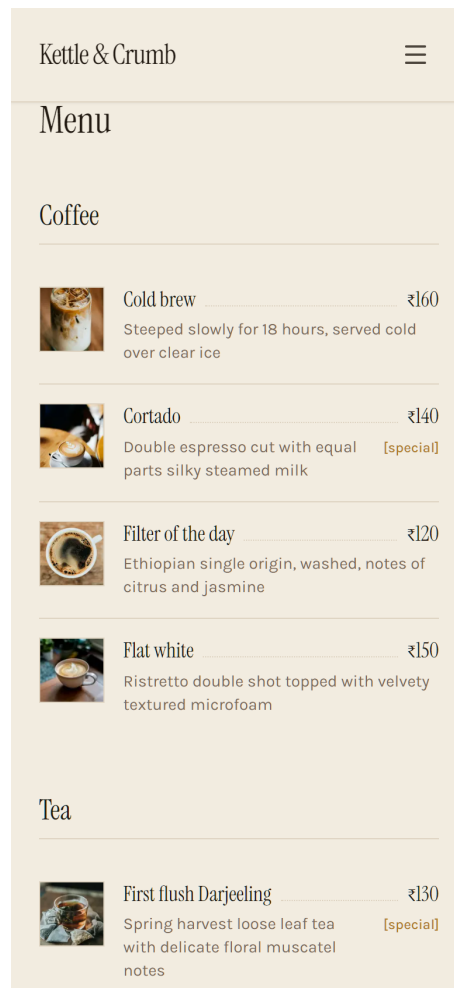
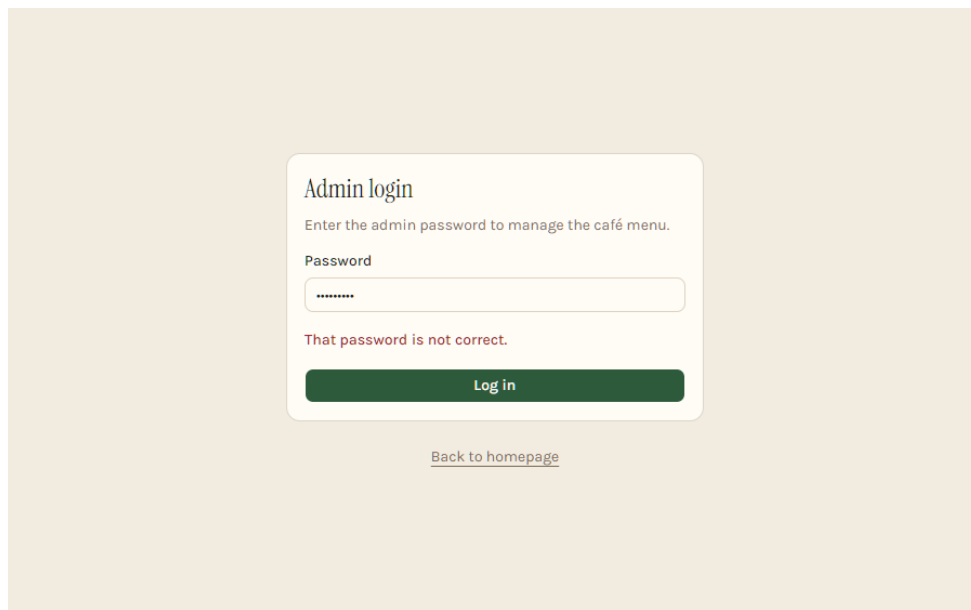


Figure D.3: Mobile View – Menu Section

4. Invalid Login



The image shows a web form titled "Admin login" on a light beige background. The form is a white rounded rectangle containing the following elements: a title "Admin login", a subtitle "Enter the admin password to manage the café menu.", a label "Password", a text input field with masked characters "*****", a red error message "That password is not correct.", a dark green "Log in" button, and a link "Back to homepage" below the form.

Admin login

Enter the admin password to manage the café menu.

Password

That password is not correct.

Log in

[Back to homepage](#)

Figure D.4: Admin Login with Incorrect Password

5. Add Item

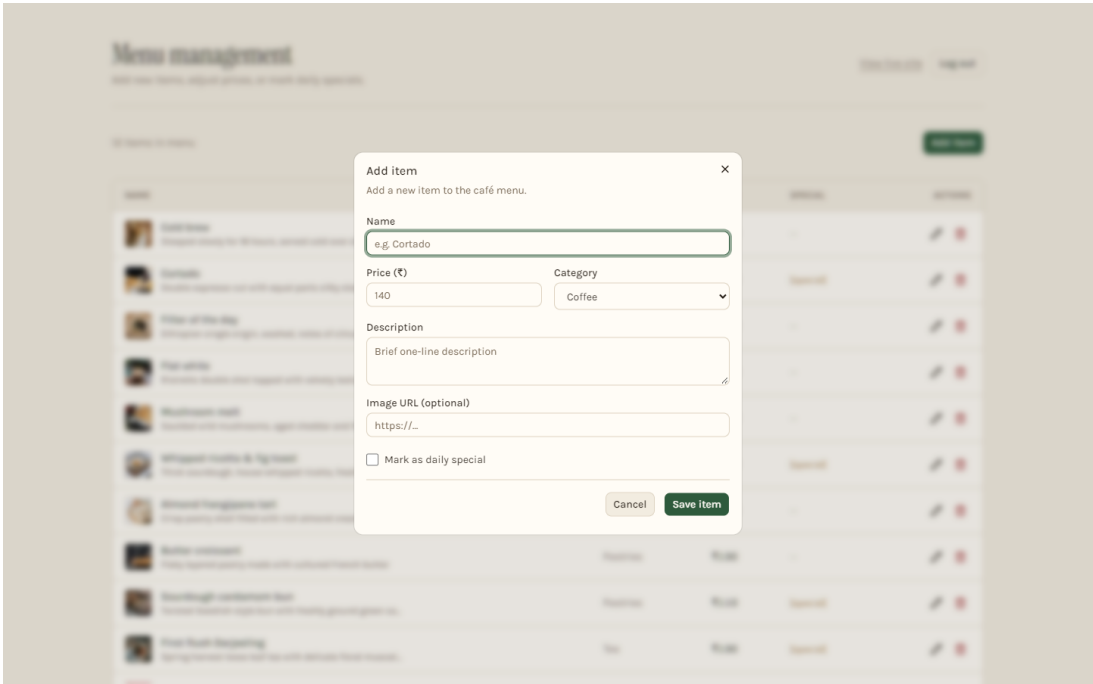


Figure D.5: Add Item Dialog